

RMO– 2014(2)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. In an acute angled triangle ABC, $\angle ABC$ is the largest angle. The perpendicular bisectors of BC and BA intersect AC at X and Y respectively. Prove that circumcentre of triangle ABC is incentre of triangle BXY.
 2. Let x, y, z be positive real numbers. Prove that $\frac{y^2 + z^2}{x} + \frac{z^2 + x^2}{y} + \frac{x^2 + y^2}{z} \geq 2(x + y + z)$.
 3. Find all pairs (x, y) of positive integers such that $2x + 7y$ divides $7x + 2y$.
 4. For any positive integer $n > 1$, let $P(n)$ denotes the largest prime not exceeding n . Let $N(n)$ denote the next prime larger than $P(n)$. (For example $P(10) = 7$ and $N(10) = 11$, while $P(11) = 11$ and $N(11) = 13$.) If $n + 1$ is a prime number, prove that the value of the sum $\frac{1}{P(2)N(2)} + \frac{1}{P(3)N(3)} + \frac{1}{P(4)N(4)} + \dots + \frac{1}{P(n)N(n)} = \frac{n-1}{2n+2}$.
 5. Let ABC be a triangle with $AB > AC$. Let P be a point on the line AB beyond A such that $AP + PC = AB$. Let M be the point mid-point of BC and let Q be the point on the side AB such that $CQ \perp AM$. Prove that $BQ = 2AP$.
 6. Suppose n is odd and each square of an $n \times n$ grid is arbitrarily filled with either by 1 or by -1. Let r_j and c_k denote the product of all numbers in the j -th row and k -th column respectively, $1 \leq j, k, \leq n$. Prove that $\sum_{j=1}^n r_j + \sum_{k=1}^n c_k \neq 0$.

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